

An Online Transfer Learning Model for Intrusion Detection using FT-Transformer and KSWIN-Driven Concept Drift Detection Mechanism

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Abstract—Deep learning has been successful in intrusion detection, but new attack methods cause data distribution changes, known as concept drift. To address this, we propose an online transfer learning method. It includes a concept drift detection mechanism to identify traffic pattern changes and fine-tune deep learning models accordingly. Experimental results show an average accuracy of 99.46% on the CIC-IDS dataset and 96.79% on the NetFlow V2 dataset.

Keywords—Network intrusion detection; Deep learning; Data stream; Concept drift detection; Transfer learning

I. INTRODUCTION

In recent research, deep learning has become increasingly prevalent in the design of highly accurate neural network models. However, existing network intrusion detection models based on deep learning often assume a static distribution of network traffic data. However, they are not applicable in dynamic network environments where malicious activities exhibit diverse and continuously evolving characteristics [1]. These elements inevitably result in alterations to the statistical properties of network traffic, an occurrence commonly referred to as concept drift [2], thereby degrading the performance of deep learning-based network intrusion detection models. We propose an Online Transfer Learning Model based on FT-Transformer and KSWIN-guided Concept Drift Detection (OTL-FT-KCD) to address these challenges. This approach combines the robust feature extraction capabilities of the FT-Transformer [3] with the online migration learning method of the KS-based concept drift detection algorithm. The primary contributions of this paper can be summarized as follows:

1. We introduce a novel concept drift detection mechanism based on KSWIN, termed KCD, to effectively address the concept drift detection problem.
2. We develop an online migration learning framework, OTL-FT-KCD, which integrates the FT-Transformer architecture and KCD, tailored explicitly for network intrusion detection scenarios.

The rest of the paper is organised, with related work discussed in the next section. We then discuss the proposed online transfer learning framework and concept drift detection mechanism and give experimental results. The last section concludes the article.

II. RELATED WORK

A. Network Intrusion Detection

A Researchers are increasingly leveraging deep learning to analyse streaming network traffic data to develop intrusion detection models. For instance, Dušan Sovilj et al. [4] employed recurrent neural networks to create unsupervised autoencoders. Based on the understanding of normal traffic distribution, this model can identify data points that cannot be effectively reconstructed. Giuseppina Andresini et al. [5] utilized the Page Hinkley detection algorithm to monitor network traffic data and detect concept drift continuously. Upon detecting significant deviations from past data in traffic, the model triggers a fine-tuning process. Li Yan et al. [6] proposed an adaptive IoT stream data analysis framework based on optimized LightGBM and concept drift adaptation. This method utilizes an optimized adaptive sliding window (OASW) to detect data drift and the PSO optimization algorithm to update the LightGBM model when data drift is detected. Gomes et al. proposed Adaptive Random Forest (ARF) [7] and Streaming Random Patches (SRP) [8], both utilizing ADWIN as a drift detector. SRP and ARF are data stream processing methods based on random forests, constructing multiple small random forests by sampling data from the data stream at each time step to achieve model updates. Mourtada et al. proposed a random forest-based AMF algorithm [9], which uses a variant of the context tree weighting algorithm to effectively perform precise aggregation on all tree prunings, obtaining an accurate online parameter-free algorithm for online detection of data streams.

B. Data Drift Detection

Online migratory learning is closely intertwined with data drift detection, particularly in scenarios where models must adjust to varying data distributions, especially in dynamic environments. Online migratory learning focuses on continuously updating the model as it receives new data streams, enabling quick adaptation to new data distributions or domain changes. There are two primary categories of data drift detection methods: statistically based methods and methods based on the performance of classification models [10].

Statistically based methods employ statistical tests to quantify the deviation between newly acquired data and established historical benchmark data when addressing concept drift. This analysis typically involves comparing the variability between feature distributions using statistical tools such as the

Chi-squared and Kolmogorov-Smirnov (KS) tests. Additionally, distance measures such as F-scatter between different probability distributions can be calculated, along with hypothesis testing theory, to reveal the dynamic evolution of data distribution over time. Standard statistical drift detection algorithms include Adaptive Windowing (ADWIN) [11], Kernel-based Change Detection with Sliding Window (KSWIN) [12], and the Page-Hinkley Test [13]. These methods uniquely monitor data flow to detect and promptly respond to potential concept drift.

Another approach is based on classification models, which detect and localize data drift using machine learning or deep learning models. This method involves training specific classifiers to distinguish between old and new datasets and monitoring changes in model performance on new datasets. Significant fluctuations in performance metrics (e.g., precision, recall, or AUC value) are considered early warning signals of potential data drift. This approach enables the timely detection and resolution of concept drift issues arising from shifting data distributions.

III. OVERALL MODEL ARCHITECTURE

A. Overall Architecture for Online Transfer Learning

Online transfer learning aims to achieve adaptive learning of models when processing data streams, enabling them to effectively adapt to data drift in the data stream. The components typically included in this architecture are shown in Figure 1.

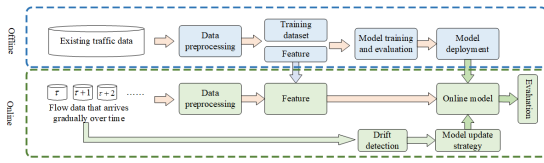


Figure 1. Overall Architecture for Online Transfer Learning

During the offline model training phase, a base model is initially trained using a substantial amount of historical or source domain data. This model is designed to perform specific tasks such as classification, regression, or prediction. Subsequently, the trained model is deployed to execute tasks in a natural online environment. Simultaneously, the model's performance is monitored through real-time data streaming, and detection techniques are employed to ascertain any concept drift or degradation in the model's performance. The model update process is activated upon detecting a change in data distribution (i.e., drift occurrence). The critical aspect of this stage is to promptly and accurately identify the model's inadequacy in processing new data.

Following model drift detection, the system dynamically adjusts and optimizes the model based on the new online data to achieve online model updating. This update may involve fine-tuning some parameters of the original model, incremental training using the new dataset or employing a transfer learning strategy to transfer knowledge from the source domain to the target domain. This adaptive process enables the model to adapt

to the new data distribution characteristics, ensuring the continued effectiveness and stability of the model's predictions.

B. KSWIN-guided Concept Drift Detection

Algorithm 1: KSWIN-guided Concept Drift Detection

Require: Data flow S , Initial classification model M , Sliding window size $win1$, Adaptive window maximum size win'_{max} , Warning threshold α , Evolutionary threshold β , Successive samples threshold n , KSWIN drift detection parameter, Amount of data for model update $win2$.

Ensure: Updated classification model M

- 1: Initialize sliding window W_i as empty, adaptive window W'_i as empty, warning status flag as False, drift status flag as False.
- 2: Initialize the KSWIN drift detector and set the parameters.
- 3: for sample i from 1 to $|S|$ do
- 4: Add $S[i]$ to the sliding window W_i and remove the oldest sample in the window if the window size exceeds $win1$.
- 5: Use the current model M to predict $S[i]$ and compute the accuracy acc_i for the sliding window W_i .
- 6: Calculating changes in feature distribution and detecting the presence of drift using the KSWIN approach.
- 7: if KSWIN detects concept drift then
- 8: Activate the drift coping mechanism, clear the adaptive window W'_i and start collecting a new subset D' for model updating.
- 9: while Collect samples until reach $win2$ do
- 10: Stop collecting and retrain the model using the samples in D' .
- 11: end while
- 12: Calculate the accuracy of the new model on D' .
- 13: Update model M to the newly trained model.
- 14: Reset warning state and drift state to False.
- 15: else
- 16: if Warning status is False and $acc_i < acc_{i-win} \times (1 - \alpha)$ then
- 17: Set the warning state to True and the drift state to False.
- 18: Add $S[i]$ to the adaptive window W'_i and record the accuracy at the start of the drift acc_i
- 19: while Warning status is True and $|W'_i| < win'_{max}$ do
- 20: Continue to add new samples to W'_i .
- 21: if Reach adaptive window maximum or the accuracy of n consecutive samples is lower than $acc_d \times (1 - \beta)$ then
- 22: Retrain the model using the samples in W'_i
- 23: Update the model M to the newly trained model.
- 24: Reset the warning state and drift state to False.
- 25: end if
- 26: end while
- 27: end if
- 28: end if
- 29: end for
- 30: Return the updated model M after completing the traversal.

KSWIN is a statistically based method that detects conceptual drift by employing the Kolmogorov-Smirnov (KS) test to compare the similarity of two sample distributions. This method uses a moving window to capture a continuous portion of the data stream and applies the KS test within the window. It divides the data stream into fixed-size windows and, for each window, compares the data distribution within the window to the reference distribution using the KS test. If the KS statistic exceeds a preset threshold, the method detects conceptual drift and updates the reference distribution to match the data distribution in the current window. While KSWIN is suitable for univariate data, it can be downscaled using Principal Component Analysis (PCA) to improve efficiency when dealing with high-

dimensional data. This paper also uses degradation in classification model performance as an indicator signal for detecting changes in data distribution. For details of the drift detection algorithm and corresponding model update strategy, refer to the step-by-step description in Algorithm 1.

The algorithm initializes the sliding window, adaptive window, and associated drift and warning status flags. For each sample in the data stream, the algorithm updates the sliding window, removes old samples based on the window size limit, and makes predictions using the current model. It calculates the prediction accuracy within the sliding window and uses the KSWIN method to detect significant changes in feature distribution (conceptual drift). Upon detecting drift, the algorithm clears the adaptive window and collects a new subset of data for model updating. When the collected data reaches a preset value, model retraining begins.

If the algorithm does not detect conceptual drift but observes a decrease in accuracy below a certain threshold, it enters a warning state and accumulates data for the adaptive window. If the window size reaches the maximum value or the accuracy of a certain number of consecutive samples is lower than the initial accuracy multiplied by a factor, the algorithm detects a possible weak drift. In this case, the algorithm retraining the model with the accumulated adaptive window data. After traversing the data stream, the algorithm returns the final classification model, considering all possible conceptual drift situations and ensuring accurate model predictions in changing data environments.

This paper proposes two adaptive model updating schemes based on drift detection to address concept drift challenges from dynamically changing data distributions. The FT-Transformer classifier is used, and an optimized update method enhances its responsiveness to potential data distribution drifts. The algorithm employs a differential training strategy during model fine-tuning: fixing some feature extraction network parameters and retraining only the classification task parameters to maintain the original pattern recognition capability while adapting quickly to emerging data features. When KSWIN detects significant changes in data distribution, the algorithm initiates a comprehensive model parameter update strategy, performing an overall retraining process to ensure the model adapts to the new data environment. Additionally, when model performance declines, the algorithm adjusts the most sensitive model hierarchy or parameter set to changes in data distribution, rather than retraining all parameters.

IV. EXPERIMENT AND ANALYSIS

A. Dataset

This study utilizes two representative datasets. The first dataset, NetFlowV2, comprises four NIDS datasets (UNSW-NB15, BoT-IoT, ToN-IoT, CSE-CIC-IDS2018), which are reextracted from the standard NetFlow-based feature set and labelled with corresponding attack categories. The second dataset is from the Canadian Institute for Cybersecurity (CIC) collaborative project, including CIC-IDS2017, CSE-CIC-IDS2018, and CIC-DDoS2019. To simulate data drift, independent sub-datasets containing 10,000 samples were randomly selected from each dataset. These sub-datasets were

merged into two new hybrid datasets to simulate data drift. This experimental design aims to thoroughly test the adaptability and effectiveness of the FT-Transformer classifier and its updating strategy in the face of changing data distributions.

B. Evaluation Indicators

In online learning environments, since data streams tend to arrive continuously and models need to be continuously updated and improved with the inflow of new data, the average accuracy over a window of time is a better indicator of the stability and adaptability of the model over time. The average accuracy rate formula is as follows:

$$\text{Average Accuracy} = \frac{1}{N} \sum_{i=1}^N \text{Accuracy}_i. \quad (1)$$

Where n denotes the number of time periods or batches, and Accuracy denotes the Accuracy_i of the i th period or batch.

C. Experimental results and analysis

To validate the effectiveness of the FT-Transformer classifier and its updating strategy in addressing data drift, we conducted a series of experiments comparing it with several commonly used online learning algorithms: SRP, ARF, and OnlineLightgbm.

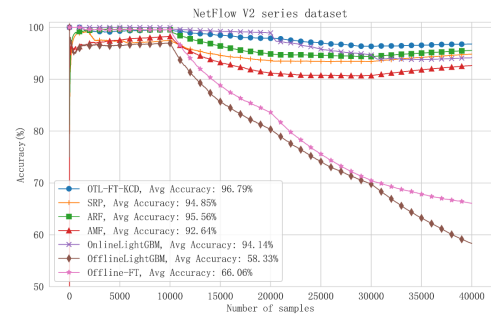


Figure 2. Netflow V2 Series Dataset Drift Adaptability Comparison

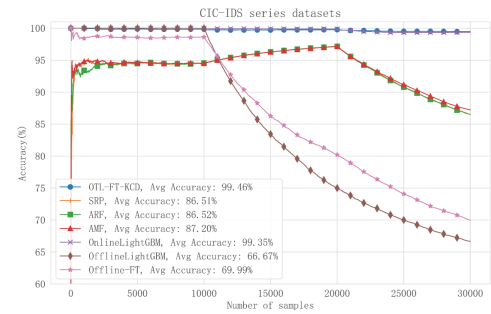


Figure 3. CIC-IDS Series Dataset Drift Adaptability Comparison

We conducted experiments on a dataset of 40,000 NetFlow V2 data samples as show in Figure 2. Each dataset comprises 10,000 consecutive data records, indicating significant data characteristic changes or distribution drifts near each integer of a thousand sample points as the model transitions between

datasets. The experimental results show a general decrease in detection accuracy for all methods near the dataset switching points, as the models need to readjust to the attack patterns and typical traffic characteristics in the new datasets. Figure 3 displays the experiment results conducted on the CIC-IDS series dataset. In contrast to other online learning methods, the OTL-FT-KCD method demonstrates robust performance and processing power when faced with the 30,000 samples and data drift between different data sources in the CIC-IDS series dataset.

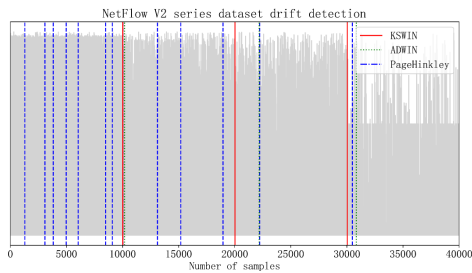


Figure 4. Netflow V2 Series Dataset Drift Detection

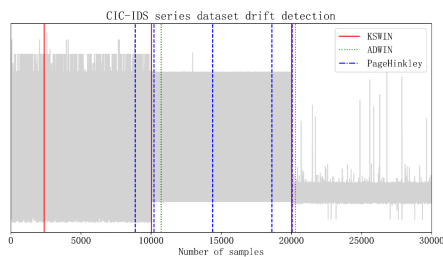


Figure 5. CIC-IDS Series Dataset Drift Detection

In order to determine the suitable techniques for detecting data drift, three typical methods, KSWIN, ADWIN and Page-Hinkley, are studied in this paper. The experimental results are presented in figures 4 and 5. The experiment shows that the KSWIN method shows higher accuracy and faster response speed in data drift detection, but it is worth noting that when processing the early part of the CIC-IDS series dataset, the false positive drift alert was undeserved.

V. CONCLUSIONS

We presents an innovative Online Transfer Learning Model (OTL-FT-KCD) designed specifically for network intrusion detection systems, which addresses the challenge posed by the inherent dynamism of network environments and the concept drift phenomenon. The proposed model incorporates a cutting-edge FT-Transformer for robust feature extraction and a novel concept drift detection mechanism guided by the KSWIN algorithm, termed KCD. This dual-component system allows for efficient identification of changes in network traffic patterns and

real-time adaptation of the intrusion detection model. Experimental evaluations showed promising results, with the OTL-FT-KCD achieving average accuracies of 99.46% on the CIC-IDS dataset and 96.79% on the NetFlow V2 dataset. Comparative studies were conducted on various drift detection techniques, including KSWIN, ADWIN, and Page-Hinkley tests. Results highlighted KSWIN's effectiveness but also noted a tendency towards false positives during the early phase of processing certain datasets.

REFERENCES

- [1] N. S. Salman and A. A. Abdulrahman, 'Survey on intrusion detection system based on analysis concept drift: status and future directions', *Int. J. Nonlinear Anal. Appl.*, vol. 14, no. 1, pp. 299–307, Jan. 2023, doi: 10.22075/ijnaa.2022.27581.3654.
- [2] S. Agrahari and A. K. Singh, 'Concept drift detection in data stream mining: a literature review', *J. King Saud Univ. Comput. Inf. Sci.*, vol. 34, no. 10, Part B, pp. 9523–9540, Nov. 2022, doi: 10.1016/j.jksuci.2021.11.006.
- [3] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, 'Revisiting Deep Learning Models for Tabular Data', presented at the Advances in Neural Information Processing Systems, Jan. 2022. Accessed: Jan. 14, 2023.
- [4] D. Sovilj, P. Budnarain, S. Sanner, G. Salmon, and M. Rao, 'A comparative evaluation of unsupervised deep architectures for intrusion detection in sequential data streams', *Expert Syst. Appl.*, vol. 159, p. 113577, Nov. 2020, doi: 10.1016/j.eswa.2020.113577.
- [5] G. Andresini, A. Appice, C. Loglisci, V. Belvedere, D. Redavid, and D. Malerba, 'A network intrusion detection system for concept drifting network traffic data', in *Discovery Science*, C. Soares and L. Torgo, Eds., Cham: Springer International Publishing, 2021, pp. 111–121. doi: 10.1007/978-3-030-88942-5_9.
- [6] L. Yang and A. Shami, 'A lightweight concept drift detection and adaptation framework for IoT data streams', *IEEE Internet Things Mag.*, vol. 4, no. 2, pp. 96–101, Jun. 2021, doi: 10.1109/IOTM.0001.2100012.
- [7] H. M. Gomes, J. Read, and A. Bifet, 'Streaming random patches for evolving data stream classification', in *2019 IEEE International Conference on Data Mining (ICDM)*, Nov. 2019, pp. 240–249. doi: 10.1109/ICDM.2019.00034.
- [8] H. M. Gomes et al., 'Adaptive random forests for evolving data stream classification', *Mach. Learn.*, vol. 106, no. 9, pp. 1469–1495, Oct. 2017, doi: 10.1007/s10994-017-5642-8.
- [9] J. Mourta, S. Gaïffas, and E. Scornet, 'AMF: aggregated mondrian forests for online learning', *J. R. Stat. Soc. Ser. B Stat. Methodol.*, vol. 83, no. 3, pp. 505–533, Jul. 2021, doi: 10.1111/rssb.12425.
- [10] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang, 'Learning under concept drift: a review', *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 12, pp. 2346–2363, Dec. 2019, doi: 10.1109/TKDE.2018.2876857.
- [11] A. Bifet and R. Gavaldà, 'Learning from time-changing data with adaptive windowing', in *Proceedings of the 2007 SIAM International Conference on Data Mining (SDM)*, in *Proceedings. , Society for Industrial and Applied Mathematics*, 2007, pp. 443–448. doi: 10.1137/1.9781611972771.42.
- [12] C. Raab, M. Heusinger, and F.-M. Schleif, 'Reactive soft prototype computing for concept drift streams', *Neurocomputing*, vol. 416, pp. 340–351, Nov. 2020, doi: 10.1016/j.neucom.2019.11.111.
- [13] R. Sebastião and J. M. Fernandes, 'Supporting the page-hinkley test with empirical mode decomposition for change detection', in *Foundations of Intelligent Systems*, M. Kryszkiewicz, A. Appice, D. Ślęzak, H. Rybinski, A. Skowron, and Z. W. Raś, Eds., Cham: Springer International Publishing, 2017, pp. 492–498. doi: 10.1007/978-3-319-60438-1.